Project Ideas

These Python-based projects provide students with practical experience in ethical hacking, focusing on essential skills such as network scanning, password cracking, and intrusion detection. Through these projects, students gain hands-on exposure to real-world cybersecurity techniques, learn key Python tools (e.g., Scapy, Hashlib), and enhance their problem-solving abilities — building a strong foundation for a career in cybersecurity. This approach effectively bridges theory and practice, preparing students to address real security challenges responsibly.

Password Hash Cracking with Hashlib

Input:

- A list of hashed passwords.
- A dictionary file containing possible plaintext passwords.
- Hashing algorithm used (e.g., MD5, SHA-256).

Output:

- Matched plaintext passwords for each hash.
- Number of attempts and time taken for the attack.

Instructions:

- **Setup**: Install any required dependencies using `pip install hashlib`.
- **Task 1**: Create a script that reads a file containing hashed passwords.
- **Task 2**: Load a dictionary file containing possible plaintext passwords.
- **Task 3:** Use the specified hashing algorithm to hash each word in the dictionary and compare it with the input hashes.
- **Task 4:** If a match is found, output the plaintext password and the corresponding hash.
- **Task 5:** Display the total number of attempts and the time taken to crack each password.

Submission Guidelines:

1. Names of Teams (group of 2) to be submitted by Nov 15th , 2024 on Discussion forum under Project Ideas module.
2. Perform your coding on the google colab file here.
3. Submit the LINK of your google colab file under Submission